MATTHEW R. SIEBERT, Ph.D. - CV

114 S Highland Ave, Baltimore, MD, 21224
603-321-5469 ◇ msiebert@stsci.edu ◇ https://msiebert1.github.io/

RESEARCH INTERESTS

- Connecting Type Ia supernova (SN Ia) physics and cosmology via spectroscopy.
- Understanding the diversity within the white dwarf supernova zoo.
- Discovering the most distant explosions and constraining the evolution in the properties of transients with cosmic time.

APPOINTMENTS

Affiliation: Space Telescope Science Institute

STScI Fellow/Postdoc - STScI *2022-present*
(50% independent science + 50% support of the HST STIS instrument team)

- Flux Recalibration Lead: Led a team to develop and release new calibration files across the STIS first-order and echelle modes, following updates to the atmospheric models of the three primary standard stars. Coordinated the fourth and fifth reference-file releases (the most complex to date) and authored two related ISRs on UV echelle calibration.
- User Support Lead: Management and training of the team of STIS Contact Scientists (CSs) and help desk fulfillers. Coordinated proposal distribution among CSs; approved program changes, coordinating with the TTRB/SPD for complex cases; oversaw the transition to reduced-gyro mode and the first "Flexible Thursday" implementations; reviewed a variety of STIS programs for detector safety.
- Science: Spectroscopic lead for the high-redshift SN effort within the Transient Science at Space Telescope group. Four first-author publications including the characterization of the first high-redshift SNe with JWST and multi-wavelength spectroscopic studies of local rare white dwarf SNe. Co-PI of an accepted Cycle 5 JWST program for high-redshift SNe Ia and led a Cycle 30 HST program to obtain STIS UV spectra of local high-velocity SNe Ia.

NSF Graduate Fellow / Graduate Research Assistant - UC Santa Cruz *2016-2022*

- PhD Thesis: *Improving Cosmological Utility of Type Ia Supernovae through Physics and Big Data* (Advisor: Ryan J. Foley).
  Science: Four first-author publications including studies on the relationship between SN Ia spectral properties and cosmological distance measurements, and progenitor scenarios of rare white dwarf SNe.

  Software: Developed an open-source SQL database and suite of analysis tools (Kaepora) for spectroscopy of SNe Ia and the UCSC spectroscopic reduction pipeline.

REU Intern - MIT Haystack Observatory *2015*

- User-interface development for the Haystack 37m radio telescope

EDUCATION

University of California Santa Cruz, Santa Cruz, CA *June 2022*
PhD: Astronomy and Astrophysics

University of California Santa Cruz, Santa Cruz, CA *December 2018*
MS: Astronomy and Astrophysics

Cornell University, Ithaca, NY *June 2016*
BS: Engineering Physics

AWARDS

GECO Postdoctoral Fellowship (UVa) ***November 2026***

STScI Achievement Award - STIS Flux Recalibration Team *2024*

STScI Postdoctoral Fellowship *2022*

Las Cumbres Observatory Postdoctoral Fellowship - *Declined* *2022*

UC Santa Cruz Dissertation Fellowship *2021*

UC Santa Cruz Department of Astronomy Whitford Prize *2018*

NSF Graduate Research Fellowship *2018*

PUBLICATIONS

First author: 8 (200 citations) [ADS], All: 114 (9300 citations), h-index: 43 [ADS]

First Author Publications

1. *SN 2025ogs: A Spectroscopically-Normal Type Ia Supernova at z = 2 as a Benchmark for Redshift Evolution*
Siebert, M. R., Pierel, J. D. R.; Engesser, M.; Coulter, D. A.; DeCoursey, C.; Fox, O. D.; Rest, A. et al. (20 authors)
ApJL Volume 1002, Issue 1, id.L3, 12 pp. 2026. [arXiv:2512.19783]

2. *Discovery of a Relativistic Stripped Envelope Type Ic-BL Supernova at z = 2.83 with JWST*
Siebert, M. R.; C. DeCoursey; D. A. Coulter; M. Engesser; J. D. R. Pierel; A. Rest; E. Egami; (22 authors)
ApJL Volume 972, Issue 1, id.L13, 13 pp. 2024. [arXiv:2406.05076]

3. *Ground-based and JWST Observations of SN 2022pul. I. Unusual Signatures of Carbon, Oxygen, and Circumstellar Interaction in a Peculiar Type Ia Supernova*
Siebert, M. R.; Kwok, L. A.; Johansson, J., ; Jha, S. W.; Blondin, S.; Dessart, L.; Foley, R. J.; Hillier, D. J.; Larison, C.; Pakmor, R.; Temim, T. (82 authors)
ApJ, Volume 960, Issue 1, id.88, 19 pp. 2024. [arXiv:2308.12449]

4. *An Asymmetric Double-degenerate Type Ia Supernova Explosion with a Surviving Companion Star*
Siebert, M. R.; Foley, R. J.; Zenati, Y.; Dimitriadis, G.; Schmidt, E.; Yang, G.; Davis,

K. W.; Taggart, K.; Rojas-Bravo, C.
ApJ, Volume 958, Issue 2, id.173, 14 pp. 2023. [arXiv:2306.11788]

5. *Strong Calcium Emission Indicates that the Ultraviolet-flashing SN Ia 2019yvq Was the Result of a Sub-Chandrasekhar-mass Double-detonation Explosion*
Siebert, M. R.; Dimitriadis, G.; Polin, A.; and Foley, R. J.
ApJL, Volume 900, Issue 2, article id.L27, 11 pp. 2020. [arXiv:2007.13793]

6. *A Possible Distance Bias for Type Ia Supernovae with Different Ejecta Velocities*
Siebert, M. R.; Foley, R. J.; Jones, D. O.; and Davis, K. W.
MNRAS, Volume 493, Issue 4, pp.5713-5725. 2020. [arXiv:2002.09490]

7. *Investigating the Diversity of Type Ia Supernova Spectra with the Open-Source Relational Database Kaepora*
Siebert, M. R.; Foley, R. J.; Jones, D. O.; Angulo, R.; Davis, K. W.; Duarte, A.; Strasburger, E.; Conlon, M.; Kazmi, N.; Nishimoto, R.; Schubert, M.; Sun, L.; and Tippens, R.
MNRAS, Volume 486, Issue 4, p.5785-5808. 2019. [arXiv:1905.02204]

8. *The Unprecedented Properties of the First Electromagnetic Counterpart to a Gravitational-wave Source*
Siebert, M. R.; Foley, R. J.; Drout, M. R.; Kilpatrick, C. D.; Shappee, B. J.; Coulter, D. A.; Kasen, D.; Madore, B. F.; Murguia-Berthier, A.; Pan, Y. -C.; Piro, A. L.; Prochaska, J. X.; Ramirez-Ruiz, E.; Rest, A.; Contreras, C.; Morrell, N.; Rojas-Bravo, C.; Simon, J. D.
ApJL, Volume 848, Issue 2, article id. L26, 6 pp. 2017. [arXiv:1710.05440]

Major Contribution Publications

9. *A spectroscopically confirmed, strongly lensed, metal-poor Type II supernova at $z = 5.13$*
Coulter, D. A.; Larison, C.; Pierel, J. D. R.; Fujimoto, S.; Kokorev, V.; Allingham, J. F. V.; Moriya, T. J.; **Siebert, M. R.** et al. (89 authors)
Submitted to Science. 2026. [arXiv:2601.04156]

10. *Discovery of a likely Type II SN at $z = 3.6$ with JWST*
Coulter, D. A.; Pierel, J. D. R.; DeCoursey, C.; Moriya, T. J.; **Siebert, M. R.** et al. (33 authors)
ApJ. Volume 1002, Issue 1, id.83, 11 pp. 2026. [arXiv:2501.05513]

11. *The New Status Qvo? SN 2021qvo is Another 2003fg-like Type Ia Supernova with a Rising Light-Curve Bump*
Paniagua, A. et al. (27 authors incl. **Siebert, M. R.**)
ApJ. Volume 997, Issue 2, id.261, 16 pp.2026. [arXiv:2508.13263]

12. *Feeling Blue: Constructing a Robust SALT3 UV Template and Constraining its Redshift Dependency*
Wang, Q.; Jones, D. O.; Pierel, J. D. R.; **Siebert, M. R.** et al. (16 authors)
Submitted to ApJ. 2025. [arXiv:2512.25064]

13. *Testing for Intrinsic Type Ia Supernova Luminosity Evolution at $z > 2$ with JWST*
Pierel, J. D. R.; Coulter, D. A.; **Siebert, M. R.** et al. (41 authors)
ApJL. Volume 981, Issue 1, id.L9, 10 pp. 2025. [arXiv:2411.11953]

14. *Discovery of an Apparent Red, High-velocity Type Ia Supernova at $z = 2.9$ with JWST*

Pierel, J. D. R.; Engesser, M.; Coulter, D. A.; DeCoursey, C.; **Siebert, M. R.** et al. (30 authors)
ApJL, Volume 971, Issue 2, id.L32, 11 pp. 2024. [arXiv:2406.05089]

15. *Ground-based and JWST Observations of SN 2022pul: II. Evidence from Nebular Spectroscopy for a Violent Merger in a Peculiar Type-Ia Supernova*
Kwok, L. A.; **Siebert, M. R.**; Johansson, J.,; Jha, S. W.; Blondin, S.; Dessart, L.; Foley, R. J.; Hillier, D. J.; Larison, C.; Pakmor, R.; Temim, T. (82 authors)
ApJ, Volume 966, Issue 1, id.135, 18 pp. 2024. [arXiv:2308.12450]

16. *Understanding Type Ia Supernova Distance Biases by Simulating Spectral Variations*
Pierel, J. D. R.; Jones, D. O.; Dai, M.; Adams, D. Q.; Kessler, R.; Rodney, S.; **Siebert, M. R.**; Foley, R. J.; Kenworthy, W. D.; Scolnic, D.
ApJ, Volume 911, Issue 2, id.96, 13 pp. 2021. [arXiv:2012.07811]

17. *Three Hypervelocity White Dwarfs in Gaia DR2: Evidence for Dynamically Driven Double degenerate Double-detonation Type Ia Supernovae*
Shen, K. J. et al. (26 authors incl. **Siebert, M. R.**)
ApJ, Volume 865, Issue 1, article id. 15, 14 pp. 2018. [arXiv:1804.11163]

Technical Publications (Unrefereed)

18. *Rederivation of STIS Secondary Echelle Mode Traces*
Siebert, M. R.; Monroe, T.; Hernandez, S.
Instrument Science Report [STIS ISR 2024-03].

19. *Recalibration of Pre-SM4 STIS Echelle Throughputs*
Siebert, M. R.; Carlberg, J. K. ; Hernandez, S. ; Monroe, T.
Instrument Science Report [STIS ISR 2024-02].

20. *Safety Acquisitions: Redundancy for non-repeatable multi-orbit STIS visits*
Dallas, M. M.; **Siebert, M. R.**
Instrument Science Report [STIS ISR 2024-01].

OBSERVING PROGRAMS

PI or Co-PI

- **Roman Space Telescope**: *2026*
Closing the Redshift Gap: Testing Type Ia Supernova Evolution Across the Cosmology Sample
(Analysis-Small, 19041, PI)

- **James Webb Space Telescope**: *2026*
Standard-ish Candles: Measuring SN Ia Luminosity Drift at $z > 3$
(121 hours, GO-12146, Co-PI)

- **Hubble Space Telescope**: *2022*
Measuring the Effect of Progenitor Metallicity on Type Ia Supernova Distance Estimates
(8 orbits, GO-17170, PI)

- **Gemini South**: *2021*
Host-Galaxy Spectroscopy of Type Ia Supernovae
(4 hours, GS-2021B-Q-243, PI)

- **Gemini South**: *2020*
  Host-Galaxy Spectroscopy of Type Ia Supernovae
  (5.8 hours, GS-2020B-Q-325, PI)

Major Contributions as Co-I

- **James Webb Space Telescope**: *2026*
  JWST Multi-Cycle Deep Transient Survey in GOODS-S
  (43 hours, GO-12577)
- **James Webb Space Telescope**: *2025*
  Constraining the Hubble Constant with Nebular-Phase Spectroscopy of a 20x Magnified, Multiply Imaged Supernova
  (7 hours, GO-12503)
- **James Webb Space Telescope**: *2025*
  SN Eos: A Multiply-Imaged, 30x Magnified SN Near the Epoch of Reionization
  (10 hours, GO-9493)
- **James Webb Space Telescope**: *2025*
  JWST Multi-Cycle Deep Transient Survey in GOODS-S
  (45 hours, GO-8060)
- **James Webb Space Telescope**: *2024*
  A Luminous, Red Transient at z = 3: An Extreme Test Bed for Supernova Evolution
  (9 hours, GO-9372)
- **James Webb Space Telescope**: *2024*
  Do Pass z=2, Do Collect Type Ia Supernovae: Breaking Out of Redshift Jail with JWST
  (76 hours, GO-5324)
- **James Webb Space Telescope**: *2023*
  The High-z Menagerie: A Rare Chance to Study the Early and Exotic Transient Universe
  (9 hours, GO-6585)
- **James Webb Space Telescope**: *2023*
  JWST NIRSpec/NIRCam Follow-Up of the High-Redshift Transients Discovered in the GOODS-S JADES-Deep Field
  (18 hours, GO-6541)
- **Hubble Space Telescope**: *2023*
  Flash Thursday: Early-time UV Spectroscopy of Type II Supernovae with HST
  (21 orbits, GO-17807)
- **Hubble Space Telescope**: *2022*
  Going Up on a Thursday: Flash Spectroscopy of Type II Supernovae with HST
  (21 orbits, GO-17507)
- **Hubble Space Telescope**: *2021*
  Measuring the Effect of Progenitor Metallicity on Type Ia Supernova Distance Estimates
  (12 orbits, GO-16690)
- **Hubble Space Telescope**: *2020*
  Measuring the Effect of Progenitor Metallicity on Type Ia Supernova Distance Estimates

(12 orbits, GO-16238)

SOFTWARE

Kaepora - An open-source relational database and suite of analysis tools for the study of SN Ia spectra (https://github.com/msiebert1/kaepora).

The UCSC Transients Spectral Reduction Pipeline - A reduction pipeline developed with supernova observations in mind for spectroscopic data from the LRIS and Kast spectrographs at Keck Observatory and Lick Observatory, respectively (https://github.com/msiebert1/UCSC_spectral_pipeline).

TECHNICAL & COMPUTER EXPERTISE

Programming: Python (numpy, scipy, astropy, matplotlib, specutils, jupyter; package development), Git/GitHub, SQL (SQLite), IDL.

HST/JWST Tools: APT, Exposure Time Calculators (STIS ETC, JWST ETC/Pandeia), stistools, CRDS.

Data Reduction: CALSTIS, JWST Calibration Pipeline, UCSC Transients Spectral Reduction Pipeline.

Space-based Observing: HST (STIS UV spectroscopy, including "Flexible Thursday" programs) and JWST (NIRSpec/NIRCam/MIRI follow-up of SNe).

Ground-based Observing: Keck (LRIS, 31 nights; AO imaging, 1 night), Shane/Lick (Kast, 22 nights), SOAR (Goodman, 7 nights), KPNO Mayall (KOSMOS, 5 nights), Nickel/Lick (imaging, 2 nights); Herbig Observational Astronomy Workshop (Lick).

SERVICE, TEACHING, AND OUTREACH

STScI Service

JWST Summer School: High Redshift Transients with JWST Instructor *2025*

STIS User Support Lead *2024-present*

STIS Flux Recalibration Working Group Lead *2023-present*

HST Cycle 32 SPG Manager *2024*

HST Cycle 31 TAC Leveler *2023*

Mentorship

Arnav Swaroop: *Spectroscopic Modeling of Type Ia Supernova Light Echos (Kaepora)* *2025*
(UC Berkeley)

Xander Jenkin: *A Comprehensive SED Library of Type Ia Supernovae for the STIS ETC (Kaepora)* *2024*
(**STScI SASP mentee**)

Matilde Mingozzi: *The Impact of STIS CCD Rotation on the E1 Wavelength Solution* *2024*
(**STIS Training Project Mentee**)

Eva Schmidt: *Strange Double-Peaked Calcium Emission in the Type Ia SN 2020hvf* *2021-2022* (Ron Ruby Scholarship)

Erika Strasburger: *Constraining the Asymmetry of Type Ia Supernovae (Kaepora)* *2019-2021* (NSF Graduate Research Fellowship, Dean's Thesis Award (UCSC), attended UC Berkeley graduate program)

Rodrigo Angulo: *Unburned carbon in the Outer Layers of Type Ia Supernova Ejecta (Kaepora)* *2018-2021* (Davidsen Fellowship (JHU), Dean's Thesis Award (UCSC), attending JHU graduate program)

Kyle Davis: *A Comprehensive Study of SN Ia Ejecta Velocities (Kaepora)* *2018-2021* (attending UCSC graduate program, postdoc at CfA-Fall 2026, active Kaepora developer)

Audrey Do: *Association of SN Ia Spectra with their Detailed Light-Curve Properties* *2018* (Kaepora)

Rey Cervantes: *Consistent Characterization of SN Ia Spectral Features (Kaepora)* *2018*

Andres Duarte: *New Techniques for the Homogenization of SN Ia Spectral Data* *2017-2018* (Kaepora)

Teaching Assistant

Introduction to Research in Physics and Astrophysics (ASTR9) *2018*

Exploring the Universe with Astronomical Data (ASTR8) *2018*

Physics of Stars (ASTR112) *2017*

Outreach

Space Astronomy Summer Program (SASP) Mentor *2024*

Lamat Program mentor *2021*

Science Internship Program (SIP) mentor *2021*

Mathematics Engineering Science Achievement (MESA) volunteer *2017-2019*

Astronomy Club Leader *2017-2019*

PRESENTATIONS

Talks

Transients in Middle Earth, Christchurch, New Zealand. *The Two Telescopes: SN Ia Redshift Evolution with Multiwavelength Spectroscopy.* February 2026.

AAS 247, Phoenix, Arizona. *SN 2025ogs: A Normal-Type Ia Supernova at z = 2.05.* January 2026.

One Hundred Years of Supernova Science, Stockholm, Sweden. *SN 2025ogs: A Normal-Type Ia Supernova at z = 2.05.* August 2025.

COSMOS Meeting 2025, Marseille, France. *The Spectroscopic Diversity of High-z Transients.* May 2025.

Transients from Space, Baltimore, MD. *Discovery of a Relativistic Stripped-envelope Type Ic-BL Supernova at z = 2.83 with JWST.* March 2025.

AAS 245, National Harbor, MD. *Discovery of a Relativistic Stripped-envelope Type Ic-BL Supernova at z = 2.83 with JWST.* January 2025.

HotSci (STScI), Baltimore, MD. *SN 2023adta: A Relativistic Stripped Envelope Supernova at z = 2.83* August 2024.

COSMOS Meeting 2024, Tokyo, Japan. *Discovery of a Relativistic Stripped-envelope Type Ic-BL Supernova at z = 2.83 with JWST.* July 2024.

AAS 243, New Orleans, LA.*An Asymmetric Double-degenerate Type Ia Supernova Explosion with a Surviving Companion Star.* January 2024.

SuperVirtual 2023. *Asymmetric Ejecta and Dust Emission in "Super-Chandrasekhar" Mass Type Ia Supernovae.* November 2023.

2023 MIAPbP Program "The Extragalactic Distance Scale and Cosmic Expansion in the Era of Large Surveys and the James Webb Telescope", Garching, Germany. *An Asymmetric Double-degenerate Type Ia Supernova Explosion with a Surviving Companion Star.* July 2023.

The Transient and Variable Universe Conference, UIUC. *An Asymmetric Double-degenerate Type Ia Supernova Explosion with a Surviving Companion Star.* June 2023.

Time Domain Astronomy Seminar, Department of Astronomy, Caltech. *Improving Cosmological Utility of Type Ia Supernovae through Physics and Big Data.* November 2021.

Roman Science Team Community Briefing, Goddard Space Flight Center, NASA. *Improving Cosmological Utility of Type Ia Supernovae through Physics and Big Data.* November 2021.

Science Happy Hour, CIERA, Northwestern University. *Improving Cosmological Utility of Type Ia Supernovae through Physics and Big Data.* November 2021.

CAPS Seminar, Center for Astrophysical Surveys, University of Illinois Urbana-Champaign. *Improving Cosmological Utility of Type Ia Supernovae through Physics and Big Data.* October 2021.

Friday Scientific Lunch Talk, NOIRLab-Tucson, NOAO. *Improving Cosmological Utility of Type Ia Supernovae through Physics and Big Data.* October 2021.

Cosmology and Astronomy Seminar, Department of Physics and Astronomy, UC Davis. *Improving Cosmological Utility of Type Ia Supernovae through Physics and Big Data.* October 2021.

Astronomy Short Talk, Department of Astronomy, UC Berkeley. *Improving Cosmological Utility of Type Ia Supernovae through Physics and Big Data.* October 2021.

Conference - Progenitors of Type Ia Supernovae, Lijiang, China. *Using a Relational Database to Investigate Spectral Diversity in a Cosmological Sample.* August 2019.

FLASH Talk, Department of Astronomy, University of Arizona. *Investigating the Diversity of Type Ia Supernova Observations with a Relational Database.* August 2018.

FLASH Talk, Department of Astronomy and Astrophysics, UC Santa Cruz. *Familiar and Exotic Events in the Current Transient Landscape.* April 2018.

Posters

AAS 246. Anchorage, AK. *"Recent Improvements to the Calibration of HST/STIS Spectroscopic Modes"*. June 2025.

Accurate Flux Calibration in the Era of Space Astronomy and All-Surveys. Baltimore, MD. *"How can we Further Improve Flux Calibration of the STIS Echelle Modes?"*. October 2024.

First Year of Science with JWST Conference. Baltimore, MD. *"JWST reveals dust emission in a Super-Chandrasekhar-mass SN Ia"*. September 2023.

Keck Science Meeting. San Diego, CA. *"The Keck Type Ia Supernova Galaxy Survey"*. September 2021.

F.O.E. Fifty-One Erg Supernova Conference. Corvallis, OR. *A Relational Database for Type Ia Supernova Observations*. June 2017.

REFERENCES

Scientific

Ori Fox: ofox@stsci.edu (STScI, 410-338-6768)

Ryan Foley: foley@ucsc.edu (University of California Santa Cruz, 831-459-2835)

Technical

Joleen Carlberg: jcarlberg@stsci.edu (STScI, 667-218-6383)